LESSON 330 (1.5 DUAL 1.3 INSTRUMENT)

Lesson Objective

This lesson reviews previous maneuvers in preparation for the End-of-Course Test.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

ATC communication procedures

- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
  - *Area of Operation III, Task A - Compliance with Air Traffic Control Clearances*
- The applicant demonstrates the ability to:
  - Correctly copy, read back, interpret, and comply with simulated or actual ATC clearances in a timely manner using standard phraseology as provided in the Aeronautical Information Manual.
  - Correctly set communication frequencies, navigation systems (identifying when appropriate), and transponder codes in compliance with the ATC clearance.
  - Use the current and appropriate paper or electronic navigation publications.
  - Intercept all courses, radials, and bearings appropriate to the procedure, route, or clearance in a timely manner.
  - Maintain the applicable airspeed ±10 knots, headings ±10°, altitude ±100 feet; and track a course, radial, or bearing within ¾-scale deflection of the CDI.

- Demonstrate SRM.
- Perform the appropriate airplane checklist items relative to the phase of flight.

VOR/DME arc

- Consider assigning 1-2 airways or radials for the student to intercept and track soon after departure.
    - For example, start with by assigning the 160 radial of the Melbourne VOR outbound within 5nm, then V3 outbound after reaching 5nm.
- Assign a DME arc from this radial or from a radar vector.
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
    - *Area of Operation V, Task A - Intercepting and Tracking Navigational Systems and Arcs*
- The applicant demonstrates the ability to:
    - Tune and correctly identify the navigation facility/program the navigation system and verify system accuracy as appropriate for the equipment installed in the airplane.
    - Determine airplane position relative to the navigational facility or waypoint.
    - Set and correctly orient to the course to be intercepted.
    - Intercept the specified course at appropriate angle, inbound to or outbound from a navigational facility or waypoint.
    - Maintain airspeed ±10 knots, altitude ±100 feet, and selected headings ±5°.
    - Apply proper correction to maintain a course, allowing no more than ¾-scale deflection of the CDI. If a DME arc is selected, maintain that arc ±1 nautical mile.
    - Recognize navigational system or facility failure, and when required, report the failure to ATC.
    - Use an MFD and other graphical navigation displays, if installed, to monitor position, track wind drift, and to maintain situational awareness.
    - Use the autopilot to make appropriate course intercepts, if installed.

Holding (choose two)

VOR
GPS
Localizer
DME/Intersection

- Assign the student at least one standard and one nonstandard hold.
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
    - *Area of Operation III, Task B - Holding Procedures*
- The applicant demonstrates the ability to:
    - Explain and use an entry procedure that ensures the airplane remains within the holding pattern airspace for a standard, nonstandard, published, or non-published holding pattern.
    - Change to the holding airspeed appropriate for the altitude or airplane when 3 minutes or less from, but prior to arriving at, the holding fix and set appropriate power as needed for fuel conservation.

- Recognize arrival at the holding fix and promptly initiate entry into the holding pattern.
- Maintain airspeed ±10 knots, altitude ±100 feet, selected headings within ±10°, and track a selected course, radial, or bearing within ¾-scale deflection of the CDI.
- Use proper wind correction procedures to maintain the desired pattern and to arrive over the fix as close as possible to a specified time and maintain pattern leg lengths when specified.
- Use an MFD and other graphical navigation displays, if installed, to monitor position in relation to the desired flightpath during holding.
- Comply with ATC reporting requirements and restrictions associated with the holding pattern.
- Demonstrate SRM.

Precision approach #1

- Tell the student to expect a precision approach.
  - Nearby examples include:
    - ILS 09R @ KMLB
    - ILS 03 or ILS 21 @ KCOF
    - ILS 36 @ KTIX
    - RNAV 09R, RNAV 09L, RNAV 27R, or RNAV 27L @ KMLB
    - RNAV 36, RNAV 09, or RNAV Z 18 @ KTIX
- Make the approach request with ATC.
- Inform the student that they are responsible for radio communications for the rest of the flight.
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
  - *Area of Operation VI, Task B - Precision Approach*
- The applicant demonstrates the ability to:
  - Accomplish the precision instrument approach(es) selected by the evaluator.
  - Establish two-way communications with ATC appropriate for the phase of flight or approach segment and use proper communication phraseology.
  - Select, tune, identify, and confirm the operational status of navigation equipment to be used for the approach.
  - Comply with all clearances issued by ATC or the evaluator.
  - Recognize if any flight instrumentation is inaccurate or inoperative and take appropriate action.
  - Advise ATC or the evaluator if unable to comply with a clearance.
  - Complete the appropriate checklist.
  - Establish the appropriate airplane configuration and airspeed considering turbulence and windshear.
  - Maintain altitude ±100 feet, selected heading ±10°, airspeed ±10 knots, and accurately track radials, courses, and bearings, prior to beginning the final approach segment.
  - Adjust the published DA/DH and visibility criteria for the aircraft approach category, as appropriate, to account for NOTAMs, Inoperative airplane or navigation equipment, or

inoperative visual aids associated with the landing environment.
    - Establish a predetermined rate of descent at the point where vertical guidance begins, which approximates that required for the airplane to follow the vertical guidance.
    - Maintain a stabilized final approach from the Final Approach Fix (FAF) to DA/DH allowing no more than ¾-scale deflection of either the vertical or lateral guidance indications and maintain the desired airspeed ±10 knots.
    - Immediately initiate the missed approach procedure when at the DA/DH, and the required visual references for the runway are not unmistakably visible and identifiable.
    - Transition to a normal landing approach (missed approach for seaplanes) only when the airplane is in a position from which a descent to a landing on the runway can be made at a normal rate of descent using normal maneuvering.
    - Maintain a stabilized visual flight path from the DA/DH to the runway aiming point where a normal landing may be accomplished within the touchdown zone.
    - Use an MFD and other graphical navigation displays, if installed, to monitor position, track wind drift, and to maintain situational awareness.

Precision approach #2

- Tell the student to expect another precision approach.
    - Nearby examples include:
        - ILS 09R @ KMLB
        - ILS 03 or ILS 21 @ KCOF
        - ILS 36 @ KTIX
        - RNAV 09R, RNAV 09L, RNAV 27R, or RNAV 27L @ KMLB
        - RNAV 36, RNAV 09, or RNAV Z 18 @ KTIX
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
    - *Area of Operation VI, Task B - Precision Approach*
- The applicant demonstrates the ability to:
    - Accomplish the precision instrument approach(es) selected by the evaluator.
    - Establish two-way communications with ATC appropriate for the phase of flight or approach segment and use proper communication phraseology.
    - Select, tune, identify, and confirm the operational status of navigation equipment to be used for the approach.
    - Comply with all clearances issued by ATC or the evaluator.
    - Recognize if any flight instrumentation is inaccurate or inoperative and take appropriate action.
    - Advise ATC or the evaluator if unable to comply with a clearance.
    - Complete the appropriate checklist.
    - Establish the appropriate airplane configuration and airspeed considering turbulence and windshear.
    - Maintain altitude ±100 feet, selected heading ±10°, airspeed ±10 knots, and accurately track radials, courses, and bearings, prior to beginning the final approach segment.

- Adjust the published DA/DH and visibility criteria for the aircraft approach category, as appropriate, to account for NOTAMs, Inoperative airplane or navigation equipment, or inoperative visual aids associated with the landing environment.
- Establish a predetermined rate of descent at the point where vertical guidance begins, which approximates that required for the airplane to follow the vertical guidance.
- Maintain a stabilized final approach from the Final Approach Fix (FAF) to DA/DH allowing no more than ¾-scale deflection of either the vertical or lateral guidance indications and maintain the desired airspeed ±10 knots.
- Immediately initiate the missed approach procedure when at the DA/DH, and the required visual references for the runway are not unmistakably visible and identifiable.
- Transition to a normal landing approach (missed approach for seaplanes) only when the airplane is in a position from which a descent to a landing on the runway can be made at a normal rate of descent using normal maneuvering.
- Maintain a stabilized visual flight path from the DA/DH to the runway aiming point where a normal landing may be accomplished within the touchdown zone.
- Use an MFD and other graphical navigation displays, if installed, to monitor position, track wind drift, and to maintain situational awareness.

Non-Precision Approach #1

- Tell the student to expect a non-precision approach.
  - Although technically any approach could be evaluated as a non-precision approach, consider assigning an approach which will not display any vertical guidance on the PFD.
  - Nearby examples which do not display vertical guidance include:
    - LOC BC 27L @ KMLB
    - VOR 09R or VOR 27L @ KMLB
    - RNAV-A or RNAV-B @ X26
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
  - *Area of Operation VI, Task A - Non-Precision Approach*
- The applicant demonstrates the ability to:
  - Accomplish the non-precision instrument approaches selected by the evaluator.
  - Establish two-way communications with ATC appropriate for the phase of flight or approach segment and use proper communication phraseology.
  - Select, tune, identify, and confirm the operational status of navigation equipment to be used for the approach.
  - Comply with all clearances issued by ATC or the evaluator.
  - Recognize if any flight instrumentation is inaccurate or inoperative and take appropriate action.
  - Advise ATC or the evaluator if unable to comply with a clearance.
  - Complete the appropriate checklist.
  - Establish the appropriate airplane configuration and airspeed considering meteorological and operating conditions.

- Maintain altitude ±100 feet, selected heading ±10°, airspeed ±10 knots, and accurately track radials, courses, and bearings, prior to beginning the final approach segment.
- Adjust the published MDA and visibility criteria for the aircraft approach category, as appropriate, for factors that include NOTAMs, inoperative aircraft or navigation equipment, or inoperative visual aids associated with the landing environment, etc.
- Establish a stabilized descent to the appropriate altitude.
- For the final approach segment, maintain no more than a ¾-scale deflection of the CDI, maintain airspeed ±10 knots, and altitude, if applicable, above MDA, +100/-0 feet, to the Visual Descent Point (VDP) or Missed Approach Point (MAP).
- Execute the missed approach procedure if the required visual references are not distinctly visible and identifiable at the appropriate point or altitude for the approach profile; or execute a normal landing from a straight-in or circling approach.
- Use an MFD and other graphical navigation displays, if installed, to monitor position, track wind drift, and to maintain situational awareness.

Non-Precision Approach #2

- Tell the student to expect a non-precision approach.
  - Although technically any approach could be evaluated as a non-precision approach, consider assigning an approach which will not display any vertical guidance on the PFD.
  - Nearby examples which do not display vertical guidance include:
    - LOC BC 27L @ KMLB
    - VOR 09R or VOR 27L @ KMLB
    - RNAV-A or RNAV-B @ X26
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
  - *Area of Operation VI, Task A - Non-Precision Approach*
- The applicant demonstrates the ability to:
  - Accomplish the non-precision instrument approaches selected by the evaluator.
  - Establish two-way communications with ATC appropriate for the phase of flight or approach segment and use proper communication phraseology.
  - Select, tune, identify, and confirm the operational status of navigation equipment to be used for the approach.
  - Comply with all clearances issued by ATC or the evaluator.
  - Recognize if any flight instrumentation is inaccurate or inoperative and take appropriate action.
  - Advise ATC or the evaluator if unable to comply with a clearance.
  - Complete the appropriate checklist.
  - Establish the appropriate airplane configuration and airspeed considering meteorological and operating conditions.
  - Maintain altitude ±100 feet, selected heading ±10°, airspeed ±10 knots, and accurately track radials, courses, and bearings, prior to beginning the final approach segment.
  - Adjust the published MDA and visibility criteria for the aircraft approach category, as

appropriate, for factors that include NOTAMs, inoperative aircraft or navigation equipment, or inoperative visual aids associated with the landing environment, etc.
    - Establish a stabilized descent to the appropriate altitude.
    - For the final approach segment, maintain no more than a ¾-scale deflection of the CDI, maintain airspeed ±10 knots, and altitude, if applicable, above MDA, +100/-0 feet, to the Visual Descent Point (VDP) or Missed Approach Point (MAP).
    - Execute the missed approach procedure if the required visual references are not distinctly visible and identifiable at the appropriate point or altitude for the approach profile; or execute a normal landing from a straight-in or circling approach.
    - Use an MFD and other graphical navigation displays, if installed, to monitor position, track wind drift, and to maintain situational awareness.

Approach with Loss of Primary Flight Indicators

- After the final approach course intercept, fail the PFD by turning down the brightness or pulling the circuit breaker.
    - NOTE: pulling the circuit breaker means the AHRS will not be operative until the aircraft comes to a stop for final alignment.
- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
    - *Area of Operation VII, Task D - Approach with Loss of Primary Flight Instrument Indicators*
- The applicant demonstrates understanding of:
    - Recognizing if primary flight instruments are inaccurate or inoperative, and advising ATC or the evaluator.
    - Common failure modes of vacuum and electric attitude instruments and how to correct or minimize the effect of their loss.
- The applicant demonstrates the ability to:
    - Advise ATC or the evaluator of if unable to comply with a clearance.
    - Complete a non-precision instrument approach without the use of the primary flight instruments using the skill elements of the non-precision approach Task (See Area of Operation VI, Task A).
    - Demonstrate SRM.

Missed Approach

- Ensure the student meets or exceeds all of the applicable Instrument ACS standards:
    - *Area of Operation VI, Task C - Missed Approach*
- The applicant demonstrates the ability to:
    - Promptly initiate the missed approach procedure and report it to ATC.
    - Apply the appropriate power setting for the flight condition and establish a pitch attitude necessary to obtain the desired performance.
    - Configure the airplane in accordance with the airplane manufacturer's instructions, establish a positive rate of climb, and accelerate to the appropriate airspeed, ±10 knots.
    - Follow the recommended checklist items appropriate to the missed approach/go-around procedure.

- Comply with the published or alternate missed approach procedure.
- Advise ATC or the evaluator if unable to comply with a clearance, restriction, or climb gradient.
- Maintain the heading, course, or bearing ±10°; and altitude(s) ±100 feet during the missed approach procedure.
- Use an MFD and other graphical navigation displays, if installed, to monitor position and track to help navigate the missed approach.
- Demonstrate SRM or CRM, as appropriate.
- Request ATC clearance to attempt another approach, proceed to the alternate airport, holding fix, or other clearance limit, as appropriate, or as directed by the evaluator.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 331
- Assign Homework:
  - Study for the mock.

Completion Standards

This lesson will be complete when the student can perform any instrument task without assistance from the instructor. The performance of each task must meet or exceed the current FAA Instrument Rating Airman Certification Standards.